Kepler's Third Law and the Flow of an Ideal Medium Around Atoms

Weight, Inertia and Body Motion Through an Ideal Medium

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Abstract

This study links Kepler's third law to the author's hypothesis of pressure-driven external flow around atoms. Atoms are described as collapsed standing waves which the non-material medium cannot enter. Their aggregates create a pressure difference, and the medium flows toward and around them, including through open toroidal centres. It has properties of an ideal incompressible fluid or gas with zero internal viscosity. A local ideal-flow comparison separates pressure-gradient force from added inertia and gives $\dot{\mathbf{v}} = \kappa D_t \mathbf{u}$, with $\kappa = (1 + c_a)/(\beta + c_a)$. At the effective condition $\beta = 1$, the response is unity. The prescribed radial profile $\mathbf{u} = -\sqrt{2E/r}\,\mathbf{e}_r$ then gives $D_t\mathbf{u} = -E\mathbf{e}_r/r^2$. A body released at rest from r_0 has $|v_r/u_r| = \sqrt{1 - r/r_0}$ without velocity damping. A separate comoving circular regime gives $\omega^2 r^3 = E$. The full vector statement also examines a possible correction $\lambda(\mathbf{u} - \mathbf{v})$ for motion with, against, across and obliquely to the flow. Rise-and-fall gravimetry provides a conditional test scale of order 10^{-3} of terrestrial support acceleration for this correction, rather than its measured value. The standing-wave condition $\beta = 1$ and the transition between the prescribed regimes remain open physical questions. The calculations distinguish these assumptions from exact field identities and classical comparison results.

Keywords: Kepler's third law, orbital parameter, ideal incompressible medium, atomic cavities, weight, inertia, relative motion, gravimetry.

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1 Introduction

1.1 The physical question

In the author's hypothesis, atoms are collapsed three-dimensional standing waves, bubble-like voids which the non-material medium cannot enter. An aggregate of atoms, a material body, creates a large pressure difference for this medium. Under its own pressure the medium flows from regions of higher pressure to regions of lower pressure, through different stages of vortex and coupled motion. It flows around atoms on all sides, including through the open centre of a torus-shaped atom. Passage through that opening is not penetration into the inaccessible atomic cavity. The medium has the properties of an ideal incompressible fluid or gas with zero internal viscosity. Where the available flow channels narrow, continuity requires an increase of local speed. The proposed interaction concerns this accelerated external flow around atoms. The formation of standing waves and the detailed conversion mechanism are outside the present study.

Earlier studies introduced the orbital parameter $E = \omega^2 r^3$ and the corresponding acceleration scale E/r^2 [1, 2, 3]. The companion work discusses matter and cumulative inflow [4], multiple systems and group approximations [5], the orbital-to-radial transition [6], and body–flow compatibility [7]. The present work connects these subjects through an explicit vector equation and observable consequences. In the present study the proposed medium is not identified with the electromagnetic field. The author intends to examine that relation in separate publications.

1.2 Results and their status

Three kinds of result are distinguished. The first consists of exact identities for specified velocity fields. The second is numerical verification of continuum calculations with given inputs. The third is a proposed physical coupling law, whose microscopic origin and empirical applicability remain to be established. In particular, reproducing E/r^2 from a chosen flow does not measure the flow or prove that atoms are cavities.

The main changes of description are substantive. Medium acceleration is retained even when body and medium velocities coincide. Relative motion is treated as a vector, including transverse motion. Weight is calibrated against support acceleration explicitly. Radial and circular fields are treated as separate regimes rather than added without a momentum balance.

2 The medium and the coupling variables

2.1 Medium properties and impermeable atomic boundaries

The ideal incompressible fluid/gas properties describe the adopted continuum comparison within its stated domain, rather than an unrestricted identification of the non-material substance with a material liquid. On an existing atomic boundary the relative normal flux is zero, $(\mathbf{u} - \mathbf{v}_{\text{wall}}) \cdot \mathbf{n} = 0$. The open centre of a torus remains an exterior flow channel [4, 7].

Write the undisturbed external velocity as $\mathbf{u}(\mathbf{x}, t)$ and the body velocity as $\mathbf{v}(t)$ at $\mathbf{x}_b(t)$. The prescribed background excludes the disturbance generated by the test body itself. Its material acceleration is

$$\mathbf{a}_{\text{m}} = D_t \mathbf{u} = \partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u}. \tag{1}$$

In the classical comparison, ρ is fluid density and V is the displaced volume. The intrinsic effective inertial coefficient is $I_0 = \beta \rho V$, and the added inertia is $c_a \rho V$. Define $I = (\beta + c_a)\rho V$. These are

comparison variables, not measured properties of the proposed medium or an assumption about celestial masses.

2.2 Two distinct rates

The possible relative-velocity correction is described by a phenomenological rate

$$\lambda_i \geq 0, \qquad [\lambda_i] = \mathrm{s}^{-1}. \tag{2}$$

It is not defined by a volume entering the atom. An impermeable ideal-flow comparison does not supply a nonzero value of this rate, and $\lambda_i = 0$ is the baseline examined here. The rate λ_m introduced below describes spatial variation of the chosen medium profile. It is not the body's coupling coefficient. No numerical value of λ_i is inferred from their kinematic relation.

3 What the medium delivers to a body

3.1 External flow and pressure forces

Incompressibility redistributes flow around an impermeable obstacle. For a sphere in a uniform potential stream, the surface speed reaches $3U/2$ at the equator. The speed increase is accompanied by a pressure decrease, but the steady pressure distribution gives zero net drag. Thus accelerated flow around an obstacle and a resultant force must be distinguished [10, 9].

The author's atomic boundary is impermeable. The force relevant to its acceleration is therefore the external pressure force, together with the unsteady response of the surrounding medium. A local pressure gradient gives a resultant even when no fluid crosses the boundary. This is the classical comparison developed next, without identifying an ordinary bubble with an atomic standing wave.

3.2 Pressure-gradient response and added inertia

In an inviscid comparison without an independently imposed bulk force, the undisturbed Euler pressure satisfies $\nabla p = -\rho \mathbf{a}_\mathrm{m}$. If its gradient varies negligibly over the displaced volume, the incident pressure force is

$$\mathbf{F}_p = -\oint_{\partial V} p\mathbf{n}\,\mathrm{d}S \simeq -V\nabla p = \rho V \mathbf{a}_\mathrm{m}. \tag{3}$$

For an isolated sphere in locally uniform potential flow, the unsteady disturbance adds $\mathbf{F}_a = c_a \rho V(\mathbf{a}_\mathrm{m} - \dot{\mathbf{v}})$, with $c_a = 1/2$. Balancing these forces with the intrinsic effective inertia gives

$$(\beta + c_a)\dot{\mathbf{v}} = (1 + c_a)\mathbf{a}_\mathrm{m}, \qquad \kappa = \frac{1 + c_a}{\beta + c_a}. \tag{4}$$

This establishes the pressure/added-inertia structure in the stated classical domain [9, 7]. For a zero-intrinsic-inertia sphere, $\beta = 0$ gives $\kappa = 3$. The author's unit-response condition is instead $\beta = 1$, which gives $\kappa = 1$ independently of the scalar c_a. An impermeable standing wave does not imply $\beta = 0$. Its effective value and applicability of the comparison remain physical hypotheses. General nonspherical forms require an added-inertia tensor. Flow gradients, circulation and interactions between atoms can add terms beyond the local sphere balance.

A steady permeable point sink obeys a different comparison, $\mathbf{F}_\mathrm{sink} = \rho q(\mathbf{u}_\mathrm{ext} - \mathbf{v})$ [8]. Its surface quadrature is retained as a classical reference, not an atomic mechanism or a definition of λ. The premise $q > 0$ through an atomic boundary is excluded here.

3.3 The complete vector hypothesis

With the effective condition $\beta = 1$, examine the body-response hypothesis

$$\boxed{\dot{\mathbf{v}} = \mathbf{a}_{\mathrm{m}} + \lambda_i(\mathbf{u} - \mathbf{v}) + \frac{\mathbf{F}_{\mathrm{ext}}}{I_i}.} \tag{5}$$

The principal term is the pressure-driven acceleration response. The possible second term describes additional interaction during body motion relative to the medium in arbitrary directions. It is not supplied by internal viscosity or by penetration into an atom. The ideal impermeable comparison has $\lambda_i = 0$. A nonzero value is a separate phenomenological extension to be tested. The external force includes support or an imposed push, and I_i is the total effective inertial coefficient. If $\beta \neq 1$, the principal term is $\kappa\mathbf{a}_{\mathrm{m}}$, and comotion in an accelerating field is not generally invariant.

The unit-response hypothesis connects weight and inertia through $I_i = (1 + c_a)\rho V$. It does not determine atomic geometry, medium density or the energy budget. Equations below use its explicit condition $\beta = 1$.

4 Supported, falling and moving bodies

4.1 Support and release

Take a short local interval, with downward unit vector $\mathbf{e}_f$, $\mathbf{u} = U\mathbf{e}_f$ and $\mathbf{a}_{\mathrm{m}} = g_{\mathrm{m}}\mathbf{e}_f$. A body held at rest requires a support opposing the acceleration

$$g_s = g_{\mathrm{m}} + \lambda U, \qquad s = \frac{\lambda U}{g_s}, \qquad g_{\mathrm{m}} = (1 - s)g_s. \tag{6}$$

For inward positive contributions, $0 \leq s \leq 1$. The support force magnitude is Ig_s. The same body immediately after release has downward acceleration g_s in this local approximation. As its velocity changes, the relative-flow contribution changes.

If $g_{\mathrm{m}} = E/R^2$ is fixed from the orbital parameter and the radial profile, then Eq. (6) predicts $g_s = E/R^2 + \lambda U$. If a trajectory simulation instead calibrates g_s to a measured value, it must use $g_{\mathrm{m}} = g_s - \lambda U$. These are different calibrations. A simulated agreement obtained by the latter choice does not independently verify the former prediction.

4.2 Free motion and exact comotion

Let $\mathbf{w} = \mathbf{v} - \mathbf{u}(\mathbf{x}_b, t)$ and set $\mathbf{F}_{\mathrm{ext}} = 0$. Differentiation along the body trajectory gives

$$\boxed{\dot{\mathbf{w}} = -\lambda\mathbf{w} - (\mathbf{w} \cdot \nabla)\mathbf{u}.} \tag{7}$$

Consequently $\mathbf{w} = 0$ is invariant wherever the field is smooth. A body already moving with the medium continues to receive $\mathbf{a}_{\mathrm{m}}$. Approach to that state depends on initial conditions and velocity gradients. A universal relaxation time $1/\lambda$ applies only to a spatially uniform flow. It cannot be transferred unchanged to a converging flow.

A freely falling body lacks a support force whether or not $\mathbf{w} = 0$. Apparent weightlessness is therefore not restricted to exact comotion. Comotion is a separate dynamical condition. Equation (7) also does not imply that an arbitrary initial slip becomes exactly zero after a finite time.

For the specified radial field and $\lambda = 0$, integration of $\ddot{r} = -E/r^2$ from rest at r_0 gives

$$v_r = -\sqrt{2E\left(\frac{1}{r} - \frac{1}{r_0}\right)}, \qquad \frac{|v_r|}{|u_r|} = \sqrt{1 - \frac{r}{r_0}}. \tag{8}$$

The ratios at $r/r_0 = 0.5, 0.1, 0.01$ are $0.707107, 0.948683, 0.994987$. Radial approach to the medium speed therefore occurs without a velocity-dependent term. It is asymptotic in the ideal point-source limit, not exact equality at finite radius. For release only 1 km above the Earth's mean radius 6371 km, the ratio at the surface is about 0.01253. This does not predict rapid comotion near a finite surface. It explains which radial initial conditions approach the flow, while transverse conditions remain separate.

4.3 Motion with, against, across and obliquely to the flow

Resolve the body velocity into $\mathbf{v} = v_\parallel \mathbf{e}_f + \mathbf{v}_\perp$. The relative-motion term is

$$\mathbf{a}_{\text{rel}} = \lambda \left[(U - v_\parallel)\mathbf{e}_f - \mathbf{v}_\perp\right]. \tag{9}$$

The comparison concerns vector velocities, not their magnitudes alone. Equal speeds in opposite directions do not remove the interaction. The four cases in Table 1 follow from the same law.

Table 1: Relative-motion response for a local flow $U\mathbf{e}_f$, with $\lambda > 0$. The medium acceleration is added separately.

Body motion	Relative-flow contribution
With the flow	Forward for $v_\parallel < U$, zero at $v_\parallel = U$ if $\mathbf{v}_\perp = 0$, backward for $v_\parallel > U$.
Against the flow	Forward contribution $\lambda(U + \lvert v_\parallel \rvert)\mathbf{e}_f$, opposing the upstream motion.
Across the flow	A contribution $\lambda U \mathbf{e}_f$ together with $-\lambda \mathbf{v}_\perp$.
Oblique motion	Both components act, with direction determined by $\mathbf{u} - \mathbf{v}$.

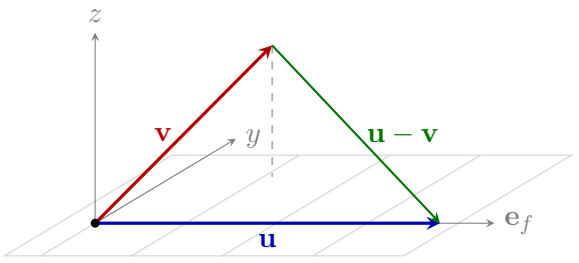


Figure 1: A three-dimensional vector construction for oblique motion. The arrow from the tip of $\mathbf{v}$ to the tip of $\mathbf{u}$ represents their difference. Multiplication by λ gives the relative-motion acceleration. The vectors are schematic, not a measured flow or an atomic geometry.

Thus the body's direction of motion is not forced to coincide with the flow direction. Its subsequent trajectory follows the vector sum in Eq. (5). A transverse component can persist or grow in a nonuniform flow even while the relative-motion term opposes it.

4.4 Pushing a body through a relatively stationary medium

The medium may move in an external frame. For a locally uniform constant velocity, choose the translating frame in which $\mathbf{u} = 0$ and $\mathbf{a}_{\text{m}} = 0$. The body velocity in this frame is the relative velocity. Then

$$\dot{\mathbf{v}} = \frac{\mathbf{F}_{\text{ext}}}{I} - \lambda \mathbf{v}. \tag{10}$$

At $\lambda = 0$, an imposed acceleration still encounters the effective inertia $I = (\beta + c_a)\rho V$. Uniform relative motion has no steady drag in the ideal comparison. A nonzero λ would add a velocity-dependent resistance and exchange energy with the medium. It is a testable extension, not a consequence of impermeability or zero viscosity. An accelerating reference frame would require its corresponding inertial terms.

4.5 Transverse motion in converging radial flow

For $\mathbf{u} = -U\mathbf{e}_r$, $U = \sqrt{2E/r}$, define $\lambda_\mathrm{m} = U/(2r)$. The local eigenvalues of $\nabla\mathbf{u}$ are $+\lambda_\mathrm{m}$ radially and $-2\lambda_\mathrm{m}$ in the two transverse directions. Small-slip rates near comotion are therefore

$$-\lambda - \lambda_\mathrm{m} \quad \text{(radial)}, \qquad -\lambda + 2\lambda_\mathrm{m} \quad \text{(transverse)}. \tag{11}$$

A positive transverse rate is compatible with angular momentum conservation under central acceleration. For the prescribed purely radial regime, the specific angular momentum obeys

$$\mathbf{h} = \mathbf{x} \times \mathbf{v}, \qquad \dot{\mathbf{h}} = -\lambda\mathbf{h}. \tag{12}$$

At $\lambda = 0$, $h = rv_t$ is conserved, so v_t grows as r decreases. Near comotion, $\dot{v}_t = (2\lambda_\mathrm{m} - \lambda)v_t$ to linear order. The rate $2\lambda_\mathrm{m}$ is not an exact statement for arbitrary radial slip. With nonzero λ, angular momentum is exchanged with the prescribed medium rather than conserved for the body alone.

5 Inverse-square acceleration and the orbital regime

5.1 A radial profile and continuity with conversion

For a steady spherically symmetric profile

$$\mathbf{u} = -\sqrt{\frac{2E}{r}}\mathbf{e}_r, \qquad D_t\mathbf{u} = -\frac{E}{r^2}\mathbf{e}_r, \qquad \lambda_\mathrm{m} = \sqrt{\frac{E}{2r^3}}, \tag{13}$$

the acceleration identity follows by differentiating $U(r)$. The signed divergence and positive volume-removal rate are

$$\nabla \cdot \mathbf{u} = -3\lambda_\mathrm{m}, \qquad \sigma = -\nabla \cdot \mathbf{u} = 3\lambda_\mathrm{m}. \tag{14}$$

This nonzero divergence needs an explicit interpretation. An incompressible fluid with no volume loss in its exterior has $\nabla \cdot \mathbf{u} = 0$. The profile in Eq. (13) instead requires a spatially averaged description with distributed loss of the flowing-medium phase. The author associates such conversion with collapse into new atoms, not entry into existing atoms. The formation process is outside this study. Its local continuity equation is $\partial_t\rho + \nabla \cdot (\rho\mathbf{u}) = -\rho\sigma$, with constant ρ. The required sink term is inferred from the prescribed profile, not independently derived from standing-wave formation. It is not the source-free microscopic flow around impermeable atoms.

The inward flux is $Q(r) = 4\pi r^2 U(r) \propto r^{3/2}$. It decreases towards the centre because of the distributed removal. A single steady point sink without distributed removal instead has $U \propto r^{-2}$ and constant Q. These two profiles cannot be interchanged.

5.2 Several sources and the prescribed Hamilton–Jacobi field

Let $W(\mathbf{x}, t) = \sum_j E_j/r_j$ and $\mathbf{u} = -\nabla\psi$. A smooth potential field satisfying

$$-\partial_t\psi + \frac{1}{2}|\nabla\psi|^2 = W + C(t) \tag{15}$$

has $D_t\mathbf{u} = \nabla W = -\sum_j E_j(\mathbf{x} - \mathbf{x}_j)/r_j^3$. This is an exact conditional identity. Additivity is specified in W, not obtained independently from atomic conversion. Its interpretation alongside component and group orbital parameters is discussed in the multiple-system study [5].

For the tested stationary two-source numerical construction on a 201×401 grid, 59,279 smooth evaluation points give median relative acceleration discrepancy 1.03% and 95th percentile 2.81%. Direct addition of two single-source velocity fields gives 61.3% and about 102% respectively. These are discrepancies within the prescribed numerical comparison, not observational accuracy. Source singularities and the nonsmooth joining region are excluded. Moving sources require the time-dependent equation, boundary conditions and a new solution.

5.3 A separate circular regime

In an equatorial circular regime with no radial flow, prescribe

$$\mathbf{u} = \sqrt{\frac{E}{r}}\mathbf{e}_\theta, \qquad D_t\mathbf{u} = -\frac{E}{r^2}\mathbf{e}_r. \tag{16}$$

Under exact comotion, Eq. (5) then gives the circular acceleration and

$$\omega^2 r^3 = E, \qquad T^2 = \frac{4\pi^2 r^3}{E}. \tag{17}$$

This verifies a circular kinematic regime under the coupling hypothesis. It does not determine its circulation or formation. Adding the radial field of Eq. (13) directly to Eq. (16) produces radial material acceleration $-2E/r^2$ and a tangential component. Such a sum does not establish the circular result. The transition needs its own field and balance [6, 7].

6 The chain of rates, acceleration and velocity

6.1 Dimensions and exact radial identities

For the specified radial profile,

$$\frac{E}{r^3} = 2\lambda_{\mathrm{m}}^2, \qquad \frac{E}{r^2} = \lambda_{\mathrm{m}} U, \qquad \frac{E}{r} = \frac{U^2}{2}. \tag{18}$$

The three quantities have units s^{-2}, $\mathrm{m\,s}^{-2}$ and $\mathrm{m}^2\,\mathrm{s}^{-2}$. The first is a squared rate, not a speed or a volume flux. The second is acceleration. The third is a squared-speed scale. Their different spatial powers are connected by the chosen profile, but their difference alone does not supply a force law. The body response is specified separately in Eq. (5).

6.2 A per-body numerical catalogue

Table 2 applies Eq. (18) separately to ten bodies. It uses the frozen orbital-parameter catalogue of the companion calculations [6, 16]. Planet-only rounded parameters are retained where used, rather than substituted with planetary-system parameters. Radii are equatorial where available. The result E/R^2 excludes rotation, nonspherical structure and local topography. The speeds and rates are model values, not measured medium properties. The Earth's equatorial catalogue value $9.798\ \mathrm{m\,s}^{-2}$ and the rounded terrestrial test value $9.8\ \mathrm{m\,s}^{-2}$ serve different purposes. The table does not assign an inflow to every celestial body as an observed fact. It gives the consequence if that body possesses the gravitational inflow postulated by the author.

Table 2: Surface values from prescribed E and R. U is the radial medium speed, distinct from orbital speed.

Body	E/R^2 ($\mathrm{m\,s^{-2}}$)	U ($\mathrm{km\,s^{-1}}$)	λ_{m} ($\mathrm{s^{-1}}$)	E/R^3 ($\mathrm{s^{-2}}$)
Sun	274.200	617.675	4.439e-04	3.941e-07
Mercury	3.699	4.249	8.705e-04	1.516e-06
Venus	8.870	10.361	8.561e-04	1.466e-06
Earth	9.798	11.180	8.764e-04	1.536e-06
Mars	3.713	5.022	7.394e-04	1.093e-06
Jupiter	24.787	59.532	4.164e-04	3.467e-07
Saturn	10.443	35.479	2.943e-04	1.733e-07
Uranus	8.869	21.293	4.165e-04	3.470e-07
Neptune	11.146	23.495	4.744e-04	4.501e-07
Moon	1.623	2.375	6.833e-04	9.337e-07

6.3 Cumulative contraction and a stationary radial profile

For a spherical moving cavity boundary $R(t)$ in a classical incompressible exterior, instantaneous continuity gives $u_r(r,t) = R^2\dot{R}/r^2$. The exterior kinetic energy is

$$K = 2\pi\rho R^3\dot{R}^2. \tag{19}$$

Only if this energy is conserved, with no net external work or energy loss, does $R^3\dot{R}^2$ remain constant. This is a time-dependent contraction law. At each instant, flux is constant across the exterior spheres. It is different from the stationary spatial profile in Eq. (13), where distributed removal changes the flux with r. A connection between these laws requires a conversion and energy balance [4, 6].

7 Observable trajectory tests

7.1 A short local fall

In the local downward convention, Eq. (6) gives

$$\dot{v} = g_s - \lambda v. \tag{20}$$

For release from rest and constant coefficients,

$$v = \frac{g_s}{\lambda}(1 - e^{-\lambda t}), \qquad x = g_s\left[\frac{t}{\lambda} - \frac{1 - e^{-\lambda t}}{\lambda^2}\right] = \frac{g_s t^2}{2} - \frac{g_s\lambda t^3}{6} + O(\lambda^2 t^4). \tag{21}$$

The $\lambda \to 0$ limit is regular. A real gravimeter also requires a vertical gradient, initial velocity, timing, optical corrections and instrument effects. Some of the cubic contribution can be absorbed by fitted parameters. Instrument accuracy in estimating g is therefore not, by itself, a bound on λ.

7.2 The IMGC-02 velocity-dependent observable

The IMGC-02 rise-and-fall analysis [14] estimates a coefficient φ in the acceleration correction $\Delta g = \varphi v$. In our local law the additional mechanism gives $\varphi_{\mathrm{model}} = -\lambda$. The measured coefficient contains several contributions:

$$\varphi_{\mathrm{obs}} = -\lambda + \varphi_{\mathrm{known}} + \varphi_{\mathrm{other}}. \tag{22}$$

The primary study includes residual gas, finite light speed and other instrument effects. For its upper-interferometer geometry the finite-light-speed contribution is $-3g/c$, approximately $-9.8 \times 10^{-8}\ \mathrm{s}^{-1}$. Other contributions need not all have the same sign.

The study reports coefficients below the scale $10^{-6}\ \mathrm{s}^{-1}$. For an example trajectory with $T = 0.3039$ s, $H = 0.1132$ m and initial speed $1.866\ \mathrm{m\,s}^{-1}$, the nonlinear analysis gives $+6.0 \times 10^{-7}\ \mathrm{s}^{-1}$ with reported standard uncertainty $2.3 \times 10^{-7}\ \mathrm{s}^{-1}$. The linear branch analysis gives $-3.788 \times 10^{-7}\ \mathrm{s}^{-1}$. The difference in estimates and signs prevents identifying either number unconditionally with $-\lambda$.

For the idealized equally spaced distance sampling and symmetric branch weighting used in that analysis, the first time moment is $C_1 = 2T/7$. With our downward sign convention, a velocity correction gives

$$g_\uparrow - g_\downarrow = -2\varphi g C_1, \qquad \Delta g_{\mathrm{model}} = \frac{4}{7}\lambda g T. \tag{23}$$

The sign follows directly from the opposite branch velocities. At $\lambda = 10^{-6}\ \mathrm{s}^{-1}$, $g = 9.8\ \mathrm{m\,s}^{-2}$ and $T = 0.3039$ s, the model branch contrast is 170.18 μGal. This is a sensitivity calculation for the specified operator, not a reanalysis of raw data.

If the unexplained opposite-velocity component is constrained to $10^{-6}\ \mathrm{s}^{-1}$, and compensating contributions are excluded, then

$$\lambda \lesssim 10^{-6}\ \mathrm{s}^{-1}, \qquad s = \frac{\lambda U}{g_s} \lesssim 1.14 \times 10^{-3} \tag{24}$$

for $U = 11186\ \mathrm{m\,s}^{-1}$ and $g_s = 9.8\ \mathrm{m\,s}^{-2}$. In that conditional interpretation, more than about 99.88% of support acceleration comes from g_{m}. This is a candidate test scale, not a confidence bound or an established decomposition of weight. Simply leaving all known contributions unsubtracted does not guarantee a conservative bound when cancellation is possible. A dedicated fit must distinguish λ from those contributions.

7.3 Comparison results and synthetic controls

The BIPM CCM.G-K2 report lists an IMGC02 degree of equivalence of -1.4 μGal with expanded uncertainty 11.1 μGal in its Table 11 [15]. These are comparison results for gravity determination, not direct bounds on the coefficient in Eq. (22). No confidence limit on s is obtained from them here.

A separate synthetic calculation calibrates the initial support acceleration to $9.8\ \mathrm{m\,s}^{-2}$ and compares quadratic fits to a 0.2 s drop and a 0.4 s rise-and-fall trajectory with initial upward speed $1.96\ \mathrm{m\,s}^{-1}$. A common vertical gradient is included. Its results are:

Chosen s	Drop minus rise-and-fall estimate (μGal)
0	approximately 0
10^{-4}	-8.59
10^{-3}	-85.86
1	-85840.14

Changing the observation windows changes the sensitivity. These numbers demonstrate an operator-dependent synthetic effect. They are not the parameters or results of actual BIPM instruments, and do not establish a 95% limit on the interaction.

8 Physical closure and open questions

8.1 Universality, inertia and the energy balance

Two test bodies at the same position and velocity have, under the assumed unit medium response,

$$\mathbf{a}_A - \mathbf{a}_B = (\lambda_A - \lambda_B)(\mathbf{u} - \mathbf{v}). \tag{25}$$

Universal initial acceleration therefore requires a universal effective coupling or an experimentally negligible difference. The observed tests of composition-independent free fall motivate this requirement [11]. Their sensitivities cannot be applied numerically to Eq. (25) without modelling the actual experiment and the local medium field.

If the coefficient of $\mathbf{a}_{\mathrm{m}}$ also varies with body composition, another differential term appears. Neither coefficient has been derived from atomic geometry. Likewise, a nonzero relative-motion term exchanges angular momentum and energy with the medium. A zero-viscosity assumption does not supply the missing conversion, pressure and energy budgets. The common origin of weight and inertia remains a physical proposal supported here by explicit consistency conditions, rather than a microscopic derivation.

8.2 Connecting the regimes

As a limited planar example, a stationary potential-flow family with circulation can be written

$$u_\theta = \frac{\ell}{r}, \qquad u_r = -\sqrt{\frac{2E}{r} + 2\varepsilon - \frac{\ell^2}{r^2}}. \tag{26}$$

Where the radicand is positive and away from the vortex singularity, this planar field has $D_t\mathbf{u} = -E\mathbf{e}_r/r^2$ and zero tangential material acceleration. Numerical finite differences in the tested region give maximum relative radial discrepancy 5.2×10^{-8} and maximum tangential acceleration 6.4×10^{-7} in the adopted units. The field is a conditional mathematical example, not a derived global three-dimensional conversion flow.

The Keplerian circular profile $u_\theta \propto r^{-1/2}$ does not belong to this family. A physical connection between circular and radial regimes needs its own continuity, circulation and conversion conditions. Another unresolved issue is whether the orbital value assigned to E already contains a relative-motion contribution. Support calibration and orbit calibration must be checked together, not adjusted independently to ensure agreement.

Finite cavity size, body back reaction and boundaries also matter. A prescribed test-body field does not guarantee conservation for the combined bodies and medium. These requirements delineate what is needed to turn the present coupling hypothesis into a closed physical model.

9 Conclusions

The physical picture. We propose that aggregates of atoms create a pressure difference in the non-material medium. Under its own pressure the medium flows toward them and around their impermeable standing-wave cavities, including through open toroidal centres, and accelerates around each atom.

Weight and inertia from one coupling. In the local potential-flow comparison, we show that the pressure-gradient force of the accelerating medium and its added inertia give one response coefficient $\kappa = (1 + c_a)/(\beta + c_a)$. At the standing-wave condition $\beta = 1$ the response is exactly unity, and one coefficient governs both the support reaction and the response to an imposed acceleration: weight and inertia arise from the same coupling.

Inverse-square acceleration from the flow. For inflow $\mathbf{u} = -\sqrt{2E/r}\,\mathbf{e}_r$ the medium's own acceleration is $-E\mathbf{e}_r/r^2$. The chain $E/r^3 = 2\lambda_{\mathrm{m}}^2$, $E/r^2 = \lambda_{\mathrm{m}}U$, $E/r = U^2/2$ connects the three quantities and gives per-body values for ten catalogued bodies. At $\beta = 1$ and $\lambda = 0$, a body released at rest in this prescribed radial field approaches the medium speed as $\sqrt{1 - r/r_0}$, and exact comotion is invariant wherever the field is smooth. A comoving circular regime gives Kepler's third law. For several sources, a potential flow obeying Eq. (15) gives additive inverse-square accelerations.

A distinguishing test. A possible additional term $\lambda(\mathbf{u} - \mathbf{v})$ is testable with rise-and-fall gravimeters, which measure the velocity dependence of acceleration directly. For $|\lambda| = 10^{-6}$ s^{-1} and the stated terrestrial profile, its contribution has the conditional scale $|\lambda|U/g_s \simeq 1.14 \times 10^{-3}$. An empirical bound requires separating instrument contributions and excluding compensating effects in the IMGC-02 analysis.

Next step. The value $\beta = 1$ for the atomic standing wave, the law of collapse of the medium into new atoms, the transition between radial and circular regimes and the energy balance remain to be derived.

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AI-based tools assisted with calculations, code development and manuscript preparation. The author is responsible for the final content.

Data and code availability

The prescribed-field checks, per-body inputs, synthetic controls and interactive visualization accompany the shared calculations [12]. The earlier visualization is archived on Zenodo [13]. Programme website: Ew2R3.org. Model trajectories are distinguished from observations and published instrument results.

References

[1] D. Kuzmenko, *The Gravitational Parameter $E = \omega^2 R^3$ as the Primary Constant of a Celestial System: Surface Gravity from Orbital Observations Without Masses or G*. ResearchGate preprint, doi:10.13140/RG.2.2.10828.37764.

[2] D. Kuzmenko, *The Orbital Invariant $E = \omega^2 a^3$ Across Scales: From the Hydrogen Atom to the Milky Way*. ResearchGate preprint, doi:10.13140/RG.2.2.24728.56327.

[3] D. Kuzmenko, *Radial Flow Interpretation of the Gravitational Invariant $E = \omega^2 R^3$: A Hydrodynamic Model for Free-Fall Acceleration Without Mass*. ResearchGate preprint, doi:10.13140/RG.2.2.33012.31364.

[4] D. Kuzmenko, *Kepler's Third Law and the Non-Material Medium: Cumulative Inflow, Atomic Cavities and a Common Origin of Weight and Inertia*. ResearchGate preprint (2026) doi:10.13140/RG.2.2.34693.10721.

[5] D. Kuzmenko, *Kepler's Third Law in Multiple Systems: Component Orbital Parameters and Controlled Group Approximations*. ResearchGate preprint (2026) doi:10.13140/RG.2.2.12882.72646.

[6] D. Kuzmenko, *Kepler's Third Law and the Orbital-to-Radial Flow Hypothesis: Corotation, Continuity and Moving Boundaries*. ResearchGate preprint (2026) doi:10.13140/RG.2.2.26304.49927.

[7] D. Kuzmenko, *Kepler's Third Law and Body–Flow Coupling: A Cavity Hypothesis and Conditional Compatibility Tests*. ResearchGate preprint (2026) doi:10.13140/RG.2.2.19593.61282.

[8] M. Lagally, Berechnung der Kräfte und Momente, die strömende Flüssigkeiten auf ihre Begrenzung ausüben, *Zeitschrift für Angewandte Mathematik und Mechanik* 2, 409–422 (1922). doi:10.1002/zamm.19220020601.

[9] MIT Department of Ocean Engineering, *Buoyancy Effects Due to Accelerating Flow*, Marine Hydrodynamics lecture notes (2003). https://web.mit.edu/fluids-modules/www/potential_flows/LecturesHTML/lec13/node2.html. The companion notes *Unsteady Motion – Added Mass* specify the sphere comparison and its conditions. https://web.mit.edu/fluids-modules/www/potential_flows/LecturesHTML/lec12/node1.html.

[10] C.E. Brennen, *Cavitation and Bubble Dynamics*, Oxford University Press, New York (1995), chapter 5.

[11] P. Touboul et al., MICROSCOPE Mission: Final Results of the Test of the Equivalence Principle, *Physical Review Letters* 129, 121102 (2022). doi:10.1103/PhysRevLett.129.121102, arXiv:2209.15487.

[12] D. Kuzmenko, *Ew2R3 shared calculations and interactive visualization: matter, multiple systems, orbital-to-radial transition and body–flow coupling*. Computer software and data (2026) doi:10.5281/zenodo.23627938.

[13] D. Kuzmenko, *Ew2R3.org interactive visualization:* $E = \omega^2 R^3$, software, v0.1.0-rc.3 (2026). Zenodo. Release doi:10.5281/zenodo.21762453. Project doi:10.5281/zenodo.21757888. Website: Ew2R3.org.

[14] V.D. Nagornyi, E. Biolcati and S. Svitlov, *Enabling a linear model for the IMGC-02 absolute gravimeter* (2014), arXiv:1312.7291v2. doi:10.48550/arXiv.1312.7291.

[15] O. Francis, H. Baumann and participants, *CCM.G-K2 Key Comparison: Final report*, November 2014. Official BIPM report.

[16] NASA/JPL, *Astrodynamic parameters* and *Planetary physical parameters*. https://ssd.jpl.nasa.gov/astro_par.html, https://ssd.jpl.nasa.gov/planets/phys_par.html. NASA planetary fact sheets supply the rounded planet-only parameters used in the companion catalogue. Catalogue snapshot: 2 October 2026.

[17] D. Kuzmenko, *$E{=}\omega^2 R^3$: Kepler's Third Law as the Basis of a Unified Physical Picture: Results of the Ew2R3.org Programme and a Call for Independent Verification*. ResearchGate preprint (2026). doi:10.13140/RG.2.2.24393.30561.